

Generation Interconnection Request System Impact Study Report

For Queue Project Q15-0453

Steeleville 100 kV
April 23, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0453. The Interconnection Customer proposes a 190.7 MW solar photovoltaic facility interconnecting at the Steeleville 100 kV substation in the Ohio area. The request was received on March 30, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 262 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$11,100,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Steeleville – Us Wind Power 100.0 kV: 135.3% loading (135 MVA) following the outage of the Lemyne – Steeleille 100.0 kV Line; DFAX = 100.0%.

The Steeleville – Lemoyne 100.0 kV is overloaded to 133.9% of its 137 MVA emergency rating for the loss of the Us Wind Power – Steeleville 100.0 kV line. The project contributes 100.0% DFAX (190.7 MW) to this overload.

The Leoyne – Troy Energy Llc 100.0 kV is overloaded to 118.4% of its 141 MVA emergency rating for the loss of the Steeleville – Us Wind Power 100.0 kV. The project contributes 100.0% DFAX (190.7 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Us Wind Power – Steeleville 100.0 kV	Rebuild line with higher-capacity conductor	\$6,056,000	100% (\$6,056,000)
Steeleville – Lmoyne 100.0 kV L	Rebuild line with higher-capacity conductor	\$3,070,000	100% (\$3,070,000)
Lemoyne – Troy Energy Llc 100.0 kV Line	Rebuild line with higher-capacity conductor	\$1,974,000	100% (\$1,974,000)
	Total allocated to project		\$11,100,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.